PromptForm AI – AI-Powered Intelligent Form Generation and Analytics Platform

Korat Princy Vikasbhai (Enrollment No.: 23SE02CS119)
B.Tech Computer Science & Engineering (7th Sem), School of Engineering, P P Savani University
Organization: **OCCUBIT SOLUTION** | Mentors: Mr. Zeel Shiyani (Company Mentor) | Ms. Kiraiti Bhatt (Institute Mentor)

Abstract

PromptForm AI is a full-stack SaaS platform that turns a one-sentence prompt or an uploaded document into a validated, shareable online form in seconds. A hybrid engine pairs Google Gemini with rule-based AIFormBrain checks, and the platform supports editing, quizzes, analytics and teamwork. All 50 automated tests passed.

Keywords: Artificial Intelligence · Form Generation · Next.js · Google Gemini · Full-Stack Development



Figure 1. Live landing page (Vercel deployment).

Introduction

- **Background:** admissions, intake, feedback and quizzes run on web forms, yet every field is still added by hand.
- **Problem:** a 10-field form took 15–25 min; field-type errors were common and PDF forms were re-typed.
- **Solution:** generate a typed, validated form from a prompt or document and manage responses in one place.

Objectives

- **Generate:** a complete form from a prompt in < 10 s.
- **Digitise:** fields from PDF, DOCX, XLSX and images.
- **Build:** ≥ 10 field types, visual builder, share link.
- **Assess:** graded quizzes with anti-cheat and analytics.
- **Secure:** JWT, team roles, ≥ 40 tests, cloud deployment.

Materials

Frontend	Next.js 15 · React 19 · Tailwind CSS
Backend	Express 4 · PostgreSQL · Prisma ORM
AI engine	Google Gemini API · AIFormBrain rules
Cloud	Vercel · Render · managed PostgreSQL

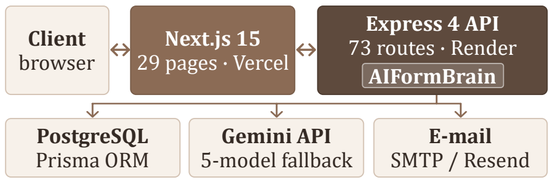


Figure 2. Three-tier system architecture.

Methodology

Agile Scrum: two-week sprints over 14 weeks, with mentor demos and four progress-report milestones.

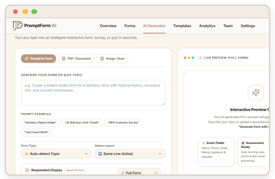
1 Problem Study	2 Requirements	3 Design
4 Core Build	5 AI Engine	6 Features
7 Testing	8 Deployment	9 Documentation

AI Form Generation Pipeline

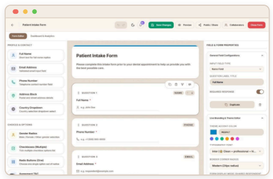


Figure 3. Hybrid generation with rule-based fallback.

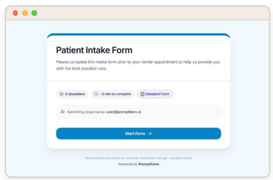
Core Modules & Features



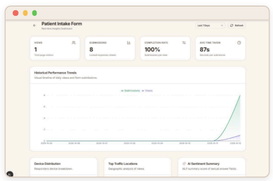
AI Form Generator
Prompt or file → form



Visual Form Builder
11 field types, themes



Responder & Quizzes
Scoring and anti-cheat



Analytics & Teams
Completion and sentiment

Results

50/50	10/10	9	≈ 8 s
Tests passed	Objectives met	Defects fixed	Prompt to form

Test ID	Scenario	Expected	Status
AUTH-03	Single active session	401	PASS
BILL-01	Free-plan 5-form limit	403	PASS
RUN-06	Tab-switch anti-cheat	201, flagged	PASS
TEAM-02	Viewer cannot edit (RBAC)	403	PASS
AI-GEN-04	Form from PDF upload	200	PASS

Table 1. Representative automated test cases.

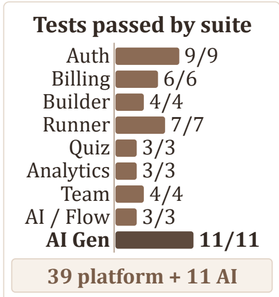


Figure 4. Passed cases per suite.

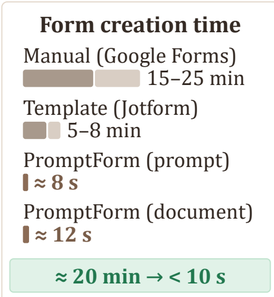


Figure 5. Time to a ready form.

Discussion

- **Faster:** creation time fell from ≈ 15–25 min to ≈ 8–12 s.
- **Reliable:** rule-based review fixed wrong field types and missing options; model fallback kept generation working during Gemini outages.
- **Secure:** API tests exposed client-only checks, which were fixed server-side and re-verified.
- **Responsive:** CRUD APIs took ≈ 40–200 ms and submissions ≈ 100–250 ms (target < 500 ms).

Scan to Visit PromptForm AI

promptform-ai-frontend.vercel.app
PromptForm AI

Conclusion

- **Summary:** a deployed, secure platform that turns a prompt or document into a shareable form in seconds; all 10 objectives were met.
- **Applications:** registrations, patient intake, surveys and classroom quizzes.
- **Limitations:** Gemini dependence, 30–50 s cold starts on free hosting, no real payments.

Recommendations

- **Payments:** Razorpay / Stripe billing.
- **Smarter AI:** generated conditional logic.
- **Multilingual:** voice-to-form, translation.
- **Integrations:** Sheets, Slack, Zapier.
- **Accessibility:** WCAG 2.2 AA audit.

Acknowledgements

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References

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